

Swapnil Das

📍 Bhubaneswar, Odisha | ✉ swapnildas018@gmail.com | 📞 +91-9337728929

PROFILE SUMMARY

Full Stack Developer specialized in Artificial Intelligence, Machine Learning, and Generative AI. Proficient in Python and Java, with hands-on experience developing AI-powered web applications using FastAPI, React, and modern machine learning frameworks. Passionate about building scalable and intelligent systems that solve real-world problems.

PROFESSIONAL EXPERIENCE

Intern — Centrox Solution Pvt. Ltd.

June 2025 – Feb 2026

- Designing and developing Generative AI and Machine Learning solutions for real-world business use cases
- Building data preprocessing, model training, and evaluation pipelines using Python and ML frameworks
- Implemented authentication and authorization using WSO2 Identity Server, Configured SSO, OAuth 2.0, OpenID Connect, and role-based access control (RBAC)
- Hands-on experience in implementation and configuration of Accops HyWorks, HySecure Gateway, and HyID.

KEY PROJECTS

QuizMaster Pro - AI-Powered Adaptive Quiz Platform

- Built Flask-based quiz platform with Google Gemini API integration for dynamic AI question generation and adaptive difficulty selection
- Implemented complete authentication system including email verification, OAuth 2.0 (Google/GitHub), and secure session management with Bcrypt
- Developed smart question recommendation algorithm with 70/30 distribution (new vs. weak areas) based on user performance tracking
- Created responsive Bootstrap 5 frontend featuring 3D flip cards, glassmorphism design, real-time quiz timer, and gamification elements (streaks, badges, leaderboard)
- Engineered admin dashboard with AI question generation, rate limiting (5/hour user, 50/day global), and analytics visualization using SQLAlchemy with optimized indexes

AI Personal Finance Dashboard - Intelligent Financial Management Platform

- Engineered FastAPI-based financial dashboard with SQLite database, serving 4 real-time analytics panels for comprehensive personal finance tracking
- Implemented AI-powered cash flow forecasting using Prophet time-series models, predicting 3-month projections with lowest point warnings and intelligent bill payment scheduling
- Developed spending velocity meter with burn rate calculation, runway estimation (days to zero balance), and period-over-period comparison analytics using scikit-learn
- Built automated subscription detection system using fuzzy pattern matching to identify recurring payments, calculate cost-per-use metrics, and generate AI cancellation recommendations via GPT-4
- Created income source diversity analyzer with concentration risk scoring, pie chart visualization, and AI-generated side hustle suggestions based on spending patterns
- Integrated Chart.js for interactive visualizations, glassmorphism UI design, and responsive grid layout supporting mobile and desktop viewports
- Implemented RESTful APIs with async/await patterns, caching layer for ML predictions (24-hour TTL), and graceful fallback to rule-based logic when OpenAI API unavailable

SCCC AI-Powered Pricing Advisor - Intelligent Cloud Cost Optimization Assistant

- Designed and developed an AI-powered pricing assistant for cloud services including ECS, OSS, and TDSQL to simplify pricing estimation and service selection
- Implemented prompt-based intelligent data extraction to dynamically collect missing user inputs and improve conversational workflow efficiency
- Built session-aware FastAPI backend to clean, validate, and process structured JSON data for generating accurate cost calculations and pricing breakdowns
- Developed cloud cost optimization logic to recommend alternative pricing configurations and improve budget planning for users

- Created responsive chat-based user interface enabling real-time pricing adjustments, interactive cost comparison, and seamless user experience
- Integrated backend validation mechanisms and fallback handling to ensure reliable pricing responses across different service configurations

TECHNICAL SKILLS

- **Languages:** Python, Core Java
- **Data Science & Analytics:** Statistical Analysis, Predictive Modeling, Feature Engineering, Data-Driven Decision Making
- **AI/ML:** Machine Learning, Deep Learning, NLP, Generative AI, Prompt Engineering, Model Training, Model Evaluation
- **Frameworks & Tools:** TensorFlow, Scikit-learn, spaCy, FastAPI, MLflow, React.js, Angular, Power BI
- **IAM & Security:** WSO2 Identity Server, Authentication, Authorization, SSO, MFA
- **Databases:** MongoDB, MySQL
- **Web Technologies:** HTML, CSS, JavaScript

EDUCATION

Bachelor of Technology in Computer Science and Engineering	CGPA: 7.98
Institute of Technical Education and Research (ITER), SOA University	2022 – 2026
12th Board (CBSE)	85%
DAV Public School, Chandrasekharpur	2021 – 2022
10th Board (CBSE)	92%
DAV Public School, Chandrasekharpur	2019 – 2020

ACHIEVEMENTS

- Selected for the Bhubaneswar District U-19 Cricket Team after a competitive selection process
- Represented the district in high-level cricket tournaments and team competitions
- Secured Runners-up position in the Inter-District Cricket Championship
- Demonstrated leadership, discipline, teamwork, and performance under pressure through competitive sports